Notes: Welch (JPE, 2004)
Capital Structure and Stock Returns

Contents

1 Big picture

This paper asks a simple but powerful question: do firms actively undo the mechanical effect of stock returns on their debt-equity ratios? If a firm's stock price rises, its market equity value rises and its market leverage mechanically falls. If a firm's stock price falls, its market leverage mechanically rises. A standard target-leverage view suggests that firms should issue or repurchase debt and equity to offset those movements over time.

Welch's answer is that, for large publicly traded U.S. firms, the answer is mostly **no**. Firms issue and retire securities actively, but this activity does not look like systematic rebalancing against stock-return-induced movements in leverage. As a result, stock returns explain a large share of debt-ratio dynamics over horizons from one to five years.

Core message. Observed capital structure is highly path-dependent. The dominant known force behind changes in market debt ratios is not an active managerial target adjustment, but the mechanical effect of firms' own stock returns on the market value of equity.

2 Incremental contribution of the paper

Main incremental contribution. The paper reframes capital structure dynamics by separating changes in debt ratios into a **mechanical stock-return component** and a **managerial net-issuing component**. This makes it possible to test whether firms actively rebalance toward prior or target debt ratios after stock price movements.

More specifically, the paper contributes in five ways.

1. **Builds the implied debt ratio benchmark.** The paper constructs the implied debt ratio, $IDR_{t,t+k}$, which is the debt ratio that would result if the firm did not issue or retire debt or equity and only its stock price changed.
2. **Shows limited active rebalancing.** Regressions of future actual debt ratios on starting debt ratios and implied debt ratios show that IDR has very high explanatory power, especially over one- to five-year horizons.
3. **Quantifies the relative importance of components.** Stock-return-induced equity growth explains roughly 40 percent of debt-ratio dynamics, while net issuing explains more, but much of that issuing activity is not used to offset stock-return effects.
4. **Separates issuing activity from capital-structure-relevant issuing activity.** Equity issues can be large but often occur where they do little to change leverage; long-term debt issuance is more important for debt-ratio dynamics.
5. **Reevaluates standard determinants of capital structure.** Once the stock-return component is included, many standard proxies - taxes, profitability, volatility, book-to-market, firm size, and industry leverage - have much less explanatory power for capital-structure changes.

Why this is new. Earlier capital structure research often interpreted the correlation between leverage and firm characteristics as evidence about managerial target choices. Welch shows that many such correlations can arise passively because stock returns move market equity values and therefore debt ratios.

3 Conceptual framework

3.1 Actual and implied debt ratios

The actual debt ratio is defined as

$$ADR_t = \frac{D_t}{E_t + D_t},$$

where D_t is the book value of debt and E_t is the market value of equity.

The implied debt ratio is the debt ratio that would occur if the firm issued no net debt or equity over the interval and equity changed only because of the stock return:

$$IDR_{t,t+k} = \frac{D_t}{E_t(1 + x_{t,t+k}) + D_t},$$

where $x_{t,t+k}$ is the stock price change net of dividends.

The key regression is

$$ADR_{t+k} = \alpha_0 + \alpha_1 ADR_t + \alpha_2 IDR_{t,t+k} + \varepsilon_{t+k}.$$

The two polar cases are:

- **Perfect readjustment:** $\alpha_1 = 1$ and $\alpha_2 = 0$. Future leverage is anchored by past actual leverage.
- **Perfect nonreadjustment:** $\alpha_1 = 0$ and $\alpha_2 = 1$. Future leverage is essentially the mechanical leverage implied by stock returns.

The empirical question is where firms lie between these extremes.

3.2 Debt-ratio dynamics

The accounting identity separates leverage changes into three forces:

1. stock-return-induced changes in the market value of equity;
2. net debt issuing or retirement;
3. net equity issuing, repurchases, and dividends.

The paper emphasizes that firms may issue a lot without actually offsetting stock-return-induced changes in debt ratios. For example, equity issuance by a nearly all-equity firm may be economically large but almost irrelevant for leverage.

4 Data and sample

The sample consists of large publicly traded U.S. corporations from Compustat and CRSP over 1962–2000. A firm enters the sample if its initial equity market capitalization is at least one-tenth of the S&P 500 level in that year. The one-year panel contains 60,317 firm-years; the five-year panel contains 40,080 firm-years.

The main debt measure uses the Compustat book value of debt. Equity value is the CRSP market value of equity. The main dependent variable is a market-value leverage ratio that combines book debt and market equity.

5 Main results

1. **Stock returns mechanically move debt ratios.** Firms with poor stock returns end up with much higher debt ratios, and firms with high stock returns end up with much lower debt ratios.
2. **Firms do not quickly rebalance.** In one-year regressions, the implied debt ratio coefficient is about 100 percent. Even over five years, the implied debt ratio remains far more important than the starting actual debt ratio.
3. **Issuing activity is active but not primarily rebalancing.** Firms issue and retire securities frequently, but the issuance does not strongly counteract stock returns.
4. **Long-term debt issuing is the most capital-structure-relevant issuing component.** Equity issuing is often large in dollar terms, but it explains relatively little of debt-ratio changes.
5. **Standard capital-structure variables lose explanatory power.** Once implied debt ratios and stock-return mechanics are included, variables such as profitability, taxes, volatility, and book-to-market explain little additional variation.

Bottom line. The proactive managerial component of capital structure remains hard to explain. The best-known observable component of debt-ratio dynamics is the firm's own stock-return history.

6 Interpretation

The results challenge simple target-adjustment models in which firms actively maintain a stable leverage target. They also complicate market-timing interpretations. Firms may issue securities, but their issuing activity does not look like systematic leverage correction after stock price shocks.

A useful way to read the paper is that capital structure is partly an accumulated record of past equity returns. High-return firms can look conservatively financed not because managers actively chose low leverage, but because equity appreciation reduced market leverage. Low-return firms can look highly levered because falling equity values raised leverage mechanically.

This also changes how we interpret cross-sectional correlations. If profitable, high market-to-book, or low-volatility firms have different stock-return histories, those variables may appear related to leverage even if managers are not actively choosing leverage based on those characteristics.

7 Identification and limitations

Strengths.

- The implied debt ratio is a clean mechanical benchmark that directly captures what stock returns alone would do to leverage.
- The decomposition distinguishes security issuance from issuance that actually matters for debt ratios.

- The analysis uses a long panel of large U.S. firms and examines horizons from one to ten years.
- The paper directly addresses a first-order omitted-variable problem in capital structure research: stock-return-induced equity-value changes.

Limitations.

- The analysis is not a randomized causal design; stock returns may reflect fundamentals, expectations, and shocks that also affect financing choices.
- The sample focuses on relatively large public firms, so the conclusions may not apply to small private firms or financially constrained startups.
- The results do not imply firms have no targets; they imply that adjustment toward targets is slow and weak relative to stock-return effects over the horizons studied.
- The paper explains much of debt-ratio dynamics but leaves the motives for net issuing activity largely unresolved.

Best interpretation. Capital structure is not well described as immediate or even medium-run readjustment to a fixed target. It is better described as the outcome of strong mechanical stock-return effects plus only partial, slow, and still poorly understood managerial security issuance.

8 Tables

8.1 Tables 1 and 2: sample magnitudes and stock-return sorting

TABLE 1
SELECTED DESCRIPTIVE STATISTICS

		One-Year			Five-Year		
Abbreviation	Description	Mean	Median	Standard Deviation	Mean	Median	Standard Deviation
ADR_t	Actual debt ratio	29.8	24.9	25.3			
$IDR_{t,t+k}$	Implied debt ratio	28.3	22.5	25.0	25.0	23.8	17.7
$E_t + D_t$	Market value (in \$ millions [2000])	\$3,294	\$552	\$14,179			
$Assets_t$	Total accounting assets (in \$ millions [2000])	\$4,645	\$583	\$21,259			
	Normalized by Market Value ($D+E$) and Winsorized (%)						
$TDNI_{t,t+1}$	Net debt issuing	3.7	.6	9.8	27.4	12.2	45.0
$ENI_{t,t+1}$	Net equity issuing before dividends	2.4	.2	6.2	21.0	5.0	41.1
$TDNI_{t,t+1} + ENI_{t,t+1}$	Debt and equity issuing	6.6	2.4	13.7	50.9	23.4	77.7
$DIV_{t,t+1} = (r_{t,t+1} - x_{t,t+1}) \cdot E_t$	Dividends	1.6	1.2	1.6	11.9	10.7	10.2
$ENI_{t,t+1} - DIV_{t,t+1}$	Activist equity expansion	.7	−.7	6.6	8.3	−4.3	43.6
$TDNI_{t,t+1} + ENI_{t,t+1} - DIV_{t,t+1}$	Activist total expansion	4.8	.9	14.0	38.1	10.9	78.8
$r_{t,t+1} \cdot E_t$	Total dollar return	8.8	5.5	28.8	80.8	46.3	112.4
$x_{t,t+1} \cdot E_t$	Induced equity growth	7.0	3.5	28.5	67.6	31.3	110.3

NOTE.—The sample is all reasonably large publicly traded firms, with a minimum market capitalization of \$75 million in the first year of the sample (1964), increasing to \$1.47 billion in the last year of the sample (2000). There were 60,317 firm-years in the one-year panel and 40,080 firm-years in the five-year panel. The firm size normalization is relative to the issue month and uses the book value for debt. The normalized series are then winsorized at the fifth and ninety-fifth percentiles.

Table 1. Selected descriptive statistics.

112 JOURNAL OF POLITICAL ECONOMY

TABLE 2
CORPORATE ACTIVITY, EQUITY GROWTH, AND CAPITAL STRUCTURE, CLASSIFIED BY STOCK RETURNS (Year-Adjusted and Sales-Adjusted)

A. SORT BY CALENDAR YEAR, SALES, ONE-YEAR NET STOCK RETURNS

	Sort Criterion, Net Return (t, $t+1$)									
	−55	−33	−22	−13	−5	3	11	21	37	79
1. Net debt issuing, $TDNI_{t,t+1}$	.7	.8	.8	.9	.8	.6	.7	.5	.1	.0
2. Net equity issuing $ENI_{t,t+1}$	.2	.2	.2	.1	.2	.2	.2	.3	.5	1.0
3. Dividends $DIV_{t,t+1}$	.0	.7	1.2	1.5	1.7	1.8	1.8	1.7	1.3	.5
4. Activist equity expansion (ENI − DIV)	0	−.4	−.9	−1.2	−1.3	−1.4	−1.3	−1.1	−.5	.1
5. Activist expansion (TDNI + ENI − DIV)	1.8	1.1	.9	.5	.5	.2	.4	.5	1.1	2.8
6. Induced equity growth, $x_{t,t+1} \cdot E_t$	−30	−16	−8	−2	2	6	12	20	33	63
7. Ending ADR_{t+1}	37	30	29	27	27	26	25	22	19	14
8. Starting ADR_t	22	22	22	23	25	25	26	25	23	22
9. Return-induced $IDR_{t,t+1}$	34	27	26	24	24	24	23	20	17	13

B. SORT BY CALENDAR YEAR, SALES, FIVE-YEAR NET STOCK RETURNS

	Sort Criterion, Net Return (t, $t+5$)									
	−106	−64	−37	−12	10	32	61	103	176	406
1. Net debt issuing, $TDNI_{t,t+5}$	6	9	10	12	13	14	14	15	15	16
2. Net equity issuing $ENI_{t,t+5}$	4	3	3	3	4	5	5	7	8	16
3. Dividends $DIV_{t,t+5}$	3	8	10	12	14	14	14	14	12	10
4. Activist equity expansion (ENI − DIV)	0	−3	−5	−7	−8	−7	−7	−6	−3	4
5. Activist expansion (TDNI + ENI − DIV)	9	9	8	7	9	10	10	13	18	32
6. Induced equity growth, $x_{t,t+5} \cdot E_t$	−37	−13	2	16	28	43	63	91	143	265
7. Ending ADR_{t+5}	41	34	32	30	29	28	25	22	18	13
8. Starting ADR_t	20	20	23	24	27	29	30	29	29	31
9. Return-induced $IDR_{t,t+5}$	34	23	23	21	20	19	17	14	11	8

NOTE.—All numbers are medians and are quoted as percentages. Firms are sorted first by calendar year, then by sales decile (to control for size), and then allocated to deciles on the basis of their stock return rank (within each group of 10 consecutive firms). In each panel, the first six rows are normalized by firm size in the month of issue; the last three rows are not. In panel A, there are between 5,849 and 6,118 observations per decile; in panel B, between 3,868 and 4,097. Net return means net of the S&P500.

Table 2. Corporate activity, equity growth, and capital structure sorted by stock returns.

Read:

- Table 1 shows that the average actual debt ratio is 29.8 percent and the median is 24.9 percent. Over one-year horizons, induced equity growth averages 7.0 percent, while net debt issuing averages 3.7 percent and net equity issuing averages 2.4 percent.
- Table 2 shows the mechanical force of stock returns. In the one-year stock-return sort, the worst-return group ends with a median debt ratio of 37 percent, while the best-return group ends with a median debt ratio of 14 percent. Over five years the corresponding ending ratios are 41 percent and 13 percent.
- The starting debt ratios do not explain these large ending differences. The ending leverage differences largely come from stock-return-induced changes in equity value.

8.2 Tables 3 and 4: readjustment tests and component decomposition

114 JOURNAL OF POLITICAL ECONOMY

TABLE 3

FAMA-MACBETH REGRESSIONS EXPLAINING FUTURE ACTUAL DEBT RATIOS ADR_{t+k} WITH DEBT RATIOS ADR_t AND STOCK RETURN–MODIFIED DEBT RATIOS $IDR_{t,t+k}$

Horizon k (Fama-MacBeth)	Constant	$IDR_{t,t+k}$	ADR_t	R^2 (%)	Cross-Sectional Regressions
		A. Without Intercept			
1-year		102.1 (1.4)	−.5 (1.4)	96.3	37
3-year		94.6 (2.1)	9.5 (2.1)	90.4	35
5-year		86.7 (2.8)	18.7 (2.1)	86.5	33
10-year		68.3 (4.6)	37.7 (1.8)	80.0	28
		B. With Intercept			
1-year	2.7 (.1)	101.4 (1.3)	−5.3 (1.2)	91.3	37
3-year	6.8 (.3)	94.4 (1.5)	−4.2 (1.4)	78.4	35
5-year	9.3 (.4)	86.9 (2.1)	−.5 (1.6)	70.2	33
10-year	13.8 (.6)	70.8 (3.7)	+6.9 (2.7)	56.0	28

NOTE.—The cross-sectional regression specifications are

$$ADR_{t+k} = [\alpha_0 +]\ \alpha_1 \cdot IDR_{t,t+k} + \alpha_2 \cdot ADR_t + \epsilon_{t+k}.$$

Reported coefficients and standard error estimates (in parentheses) are computed from the time series of cross-sectional regression coefficients and quoted as percentages. A coefficient of 100 percent on implied debt ratio ($IDR_{t,t+k}$) indicates perfect lack of readjustment, and a coefficient of 100 percent on actual debt ratio (ADR_t) indicates perfect readjustment. The R^2's are time-series averages of cross-sectional estimates. In the one-year regressions, 60,317 firm-years are used; in the 10-year regressions, 25,180 firm-years.

Table 3. Future actual debt ratios explained by prior actual debt ratios and implied debt ratios.

116 JOURNAL OF POLITICAL ECONOMY

TABLE 4

EXPLANATORY POWER OF COMPONENTS OF DEBT RATIOS AND DEBT RATIO DYNAMICS: TIME-SERIES AVERAGE R^2'S FROM CROSS-SECTIONAL FAMA-MACBETH REGRESSIONS

	k = 1 YEAR, AVERAGE R^2		k = 5 YEARS, AVERAGE R^2	
	Levels (%)	Differences (%)	Levels (%)	Differences (%)
1. Past debt ratio, ADR_t	85.0		54.0	
2. Implied debt ratio, $IDR_{t,t+k}$	91.2	43.2	68.3	40.2
3. Implied debt ratio, with dividend payout	91.3	43.6	70.0	41.5
4. All issuing and dividend activity	93.4	56.9	85.7	68.8
5. All issuing activity	93.4	56.5	84.9	65.9
6. Net equity issuing activity	85.6	5.2	63.3	16.2
7. Net debt issuing activity	92.3	49.7	71.3	40.6
Convertibles only		3.3		4.0
Short-term only		15.9		13.7
Long-term only		30.7		31.9

NOTE.—In levels, ADR_{t+k} is explained by the regressor. In differences, $ADR_{t+k} - ADR_t$ is explained by the regressor minus ADR_t. The ratios are defined as follows: row 1: $D_t/(D_t + E_t)$; row 2: $D_t/[D_t + E_t \cdot (1 + x_{t,t+k})]$; row 3: $D_t/[D_t + E_t \cdot (1 + r_{t,t+k})]$; row 4: $(D_t + TDNI_{t,t+k})/(D_t + TDNI_{t,t+k} + E_t + ENI_{t,t+k} - DIV_{t,t+k})$; row 5: $(D_t + TDNI_{t,t+k})/(D_t + TDNI_{t,t+k} + E_t + ENI_{t,t+k})$; row 6: $D_t/(D_t + E_t + ENI_{t,t+k} - DIV_{t,t+k})$; row 7: $(D_t + TDNI_{t,t+k})/(D_t + TDNI_{t,t+k} + E_t)$.

Table 4. Explanatory power of components of debt ratios and debt-ratio dynamics.

Read:

- Table 3 is the core readjustment test. Without an intercept, the one-year IDR coefficient is 102.1 percent and the five-year coefficient is 86.7 percent. This is close to the nonreadjustment benchmark.
- Adding an intercept lowers overall fit but does not overturn the conclusion. Over one year, IDR still has a coefficient of 101.4 percent, while the starting actual debt ratio has little role.
- Table 4 shows that IDR alone explains 43.2 percent of one-year debt-ratio differences and 40.2 percent of five-year differences. All issuing activity explains more, but net equity issuing alone explains very little; long-term debt issuing is much more relevant.

8.3 Table 5 and Appendix Table A1: standard proxies and variable definitions

TABLE 5

Fama-MacBeth Regressions Explaining Debt Ratio Changes ($ADR_{t+k} - ADR_t$): Adding Variables Used in Prior Literature

Variable	k = 1 Year (N=57,921) Multivariate	Four-Variate	Standard Deviation	k = 5 Years (N=34,880) Multivariate	Four-Variate	Standard Deviation
Intercept	2.70*	varies		−.62	varies	
	Flow Variables Measured from t to $t+k$					
$\Delta IDR \equiv IDR_{t,t+k} - ADR_t$	7.38***	varies	.07	10.47**	varies	.13
Log volatility	−2.54***	−.61***	.80	−5.27***	−.30	.67
× ΔIDR	−.97	−.18	.19	3.33	3.83*	.37
Stock return	−.06	−.28	.54	−1.33	−1.68	3.78
× ΔIDR	.91***	.74***	.01	1.76**	3.19***	.02
Tax rate	.25*	−.05	.28	.74***	.73*	.17
× ΔIDR	−.03	.22	.03	.19	1.75***	.05
	Stock Variables Measured at t					
Industry deviation	−1.83***	−1.13***	.19	−6.87***	−4.86***	.21
× ΔIDR	−.14	−.14	.01	−.91*	−.73	.03
Log assets	−2.71***	−.20*	1.88	−4.67***	−.58**	1.93
× ΔIDR	−2.62	.22	.44	1.42	−1.49	.84
Log relative market capitalization	1.77***	.01	1.54	3.47***	.54	1.71
× ΔIDR	.75	.12	.10	.53	−.07	.26
Book-market ratio	.16	−.22	.55	.75*	−1.19***	.57
× ΔIDR	.37	.24	.07	.61	−.02	.17
Cross-sectional regressions	37	varies		33	varies	
Average R^2	54%	varies		59%	varies	

Note.—The reported coefficient estimates are time-series means of cross-sectional regression coefficients (1966–2000) and are quoted as percentages. Except for the intercept, variables were unit-normalized (coefficients were multiplied by the standard deviation of the variable). The multivariate columns are the coefficients from one big specification. The four-variate columns are the coefficients from individual specifications, one variable V at a time, $ADR_{t+k} - ADR_t = \alpha_0 + \alpha_1 \cdot X_{t,t+k} + \alpha_2 \cdot V + \alpha_3 \cdot V \times X_{t,t+k} + \epsilon_{t+k}$. Four-variate regressions avoid multicollinearity.

* The Fama-MacBeth-type t-statistic is above 3.

** The Fama-MacBeth-type t-statistic is above 4.

*** The Fama-MacBeth-type t-statistic is above 5.

Table 5. Adding variables used in prior capital-structure literature.

TABLE A1

Variable Definitions

Variable	Definition
D_t	The sum of long-term debt (Compustat item [9]) and debt in current liabilities ([34]); convertible securities are excluded; broader definitions were tried and found not to affect the results
E_t	CRSP market value, the number of shares times closing price at the end of the fiscal year; exactly time-aligned with D_t
Assets	Assets, defined by [6], adjusted to 2000 levels using the CPI
ADR_t	Actual debt ratio: $D_t/(E_t + D_t)$ (the dependent variable); all results, including the relative importance of debt issuing and equity issuing, are invariant to predicting an actual equity ratio (AER) instead of ADR (because AER ≡ 1 − ADR_t) or to predicting D_t/EQ_t
$IDR_{t,t+k}$	Implied debt ratio: $D_t/[E_t \cdot (1 + x_{t,t+k}) + D_t]$
$r_{t,t+k}$	Stock returns with dividends, from CRSP
$x_{t,t+k}$	Stock returns without dividends, from CRSP
$TDNI_{t,t+k}$	Difference in total debt value: $D_{t+k} - D_t$
$ENI_{t,t+k}$	Difference in total equity value without return and dividend effects: $E_{t+k} - E_t \cdot (1 + x_{t,t+k})$
$DIV_{t,t+k}$	$(r_{t,t+k} - x_{t,t+k}) \cdot E_t$
Tax rate	Total income tax ([16]) divided by the sum of earnings ([53] × [54]) plus income tax ([16]); winsorized to between −100% and +200%
Log equity volatility	Standard deviation of returns, computed from CRSP; timed from $t-1$ to t; logged, not winsorized
Log firm volatility	Equity volatility times $E_t/(E_t + D_t)$; logged, not winsorized
Mergers and acquisitions activity	A dummy of one if Compustat footnotes indicate a flag of AA, AR, AS, FA, FB, FC, AB, FD, FE, or FF
Profitability/sales	Operating income ([13]) divided by sales ([12]); winsorized at 5% and 95%
Profitability/assets	Operating income ([13]) divided by assets ([6]); winsorized at 5% and 95%
Stock returns	As used in the IDR computation, but by itself
Book-market ratio	Book value of equity ([60]) divided by CRSP market value; winsorized at 5% and 95%
Log assets	Defined by [6], adjusted to 2000 levels using the CPI; not winsorized
Log relative market capitalization	E divided by the price level of the S&P500; always greater than 0.1
Deviation from industry debt ratio	Industry herding or conformity: ADR of a firm minus the ADR average in its three-digit SIC code industry (similar answers are obtained if two-digit industry definitions are used instead)*
Selling expense	Selling expense ([189]) divided by sales ([12]); winsorized at 5% and 95%
R&D expense	R&D expense ([46]) divided by sales ([12]); winsorized at 5% and 95%

TABLE A1

(*Continued*)

Variable	Definition
Interest coverage	Operating income ([13]) divided by interest paid ([15]); winsorized at 5% and 95%
Graham's tax rate	Provided by John Graham (see Graham 2000)
Future stock return reversals	This variable is *not* fully known at time t; it multiplies the stock returns from t, $t+k$ with stock returns from $t+k$, $t+2k$; the variable measures future price continuation (consecutive positive and consecutive negative rates of return result in positive values); not winsorized
Profitability changes	This variable is *not* fully known at time t; profitability divided by sales, an average from t to $t+k$, minus profitability divided by sales at $t-2$ (other definitions were tried and not found to make a difference); not winsorized

Note.—In tables 1 and 2, ENI, DIV, and induced equity growth were obtained from the sum of market values minus lagged market values adjusted by compounded returns. The reported statistics sit about halfway between statistics normalized by firm value at the beginning of the period (t) and statistics normalized by firm value at the end of the period ($t + k$). All independent *stock* variables used in table 5 have to be known at the outset, i.e., at time t, when predicting ADR_{t+k}. All *flow* variables are measured over the same time interval as the rate of return, i.e., from t to $t + k$. (Over five years, the reported standard deviations pertain to the average over the five flow variables, not the sum.) In some cases, various variable definitions were explored. The results were robust in that no reported insignificant variable achieved significance in alternatives, and vice versa.

* This variable was also used in Hovakimian et al. (2001). MacKay and Philips (2002) independently provide a more detailed examination of industry capital structure. Bikhchandani et al. (1998) review the herding literature.

Appendix Table A1. Variable definitions.

Read:

- Table 5 adds variables commonly used in the capital-structure literature. The implied debt-ratio term remains economically dominant.
- Several standard proxies are statistically significant in some specifications, but their economic

role is much smaller than the stock-return-induced component.

- Appendix Table A1 is useful because the interpretation depends heavily on timing. Flow variables are measured over the same interval as the return, while stock variables must be known at time t when predicting future debt ratios.

9 Takeaway for future research

This paper suggests that empirical capital-structure work should treat past stock returns and implied debt ratios as first-order controls. Otherwise, variables that correlate with past equity performance can appear to explain leverage choices even when they are mostly picking up mechanical market-value changes.

The main open question left by the paper is not whether firms issue securities - they clearly do - but why their issuing activity does not more directly offset the large leverage movements created by stock returns. A successful theory of capital structure dynamics must explain both the strength of the mechanical stock-return effect and the limited, slow, and incomplete managerial response to it.